

Bulking agents, antispasmodics and antidepressants for the treatment of irritable bowel syndrome.

BACKGROUND: Irritable bowel syndrome (IBS) is a common chronic gastrointestinal disorder. The role of pharmacotherapy for IBS is limited and focused mainly on symptom control. **OBJECTIVES:** The objective of this systematic review was to evaluate the efficacy of bulking agents, antispasmodics and antidepressants for the treatment of irritable bowel syndrome. **SEARCH STRATEGY:** Computer assisted structured searches of MEDLINE, EMBASE, The Cochrane library, CINAHL and PsychInfo were conducted for the years 1966-2009. An updated search in April 2011 identified 10 studies which will be considered for inclusion in a future update of this review. **SELECTION CRITERIA:** Randomized controlled trials comparing bulking agents, antispasmodics or antidepressants with a placebo treatment in patients with irritable bowel syndrome aged over 12 years were considered for inclusion. Only studies published as full papers were included. Studies were not excluded on the basis of language. The primary outcome had to include improvement of abdominal pain, global assessment or symptom score. **DATA COLLECTION AND ANALYSIS:** Two authors independently extracted data from the selected studies. Risk Ratios (RR) and Standardized Mean Differences (SMD) with 95% confidence intervals (CI) were calculated. A proof of practice analysis was conducted including sub-group analyses for different types of bulking agents, spasmolytic agents or antidepressant medication. This was followed by a proof of principle analysis where only the studies with adequate allocation concealment were included. **MAIN RESULTS:** A total of 56 studies (3725 patients) were included in this review. These included 12 studies of bulking agents (621 patients), 29 of antispasmodics (2333 patients), and 15 of antidepressants (922 patients). The risk of bias was low for most items. However, selection bias is unclear for many of the included studies because the methods used for randomization and allocation concealment were not described. No beneficial effect for bulking agents over placebo was found for improvement of abdominal pain (4 studies; 186 patients; SMD 0.03; 95% CI -0.34 to 0.40; $P = 0.87$), global assessment (11 studies; 565 patients; RR 1.10; 95% CI 0.91 to 1.33; $P = 0.32$) or symptom score (3 studies; 126 patients SMD -0.00; 95% CI -0.43 to 0.43; $P = 1.00$). Subgroup analyses for insoluble and soluble fibres also showed no statistically significant benefit. Separate analysis of the studies with adequate concealment of allocation did not change these results. There was a beneficial effect for antispasmodics over placebo for improvement of abdominal pain (58% of antispasmodic patients improved compared to 46% of placebo; 13 studies; 1392 patients; RR 1.32; 95% CI 1.12 to 1.55; $P < 0.001$; NNT = 7), global assessment (57% of antispasmodic patients improved compared to 39% of placebo; 22 studies; 1983 patients; RR 1.49; 95% CI 1.25 to 1.77; $P < 0.0001$; NNT = 5) and symptom score (37% of antispasmodic patients improved compared to 22% of placebo; 4 studies; 586 patients; RR 1.86; 95% CI 1.26 to 2.76; $P < 0.01$; NNT = 3). Subgroup analyses for different types of antispasmodics found statistically significant benefits for cimetopium/ dicyclomine, peppermint oil, pinaverium and trimebutine. Separate analysis of the studies with adequate allocation

concealment found a significant benefit for improvement of abdominal pain. There was a beneficial effect for antidepressants over placebo for improvement of abdominal pain (54% of antidepressants patients improved compared to 37% of placebo; 8 studies; 517 patients; RR 1.49; 95% CI 1.05 to 2.12; $P = 0.03$; NNT = 5), global assessment (59% of antidepressants patients improved compared to 39% of placebo; 11 studies; 750 patients; RR 1.57; 95% CI 1.23 to 2.00; $P < 0.001$; NNT = 4) and symptom score (53% of antidepressants patients improved compared to 26% of placebo; 3 studies; 159 patients; RR 1.99; 95% CI 1.32 to 2.99; $P = 0.001$; NNT = 4). Subgroup analyses showed a statistically significant benefit for selective serotonin releasing inhibitors (SSRIs) for improvement of global assessment and for tricyclic antidepressants (TCAs) for improvement of abdominal pain and symptom score. Separate analysis of studies with adequate allocation concealment found a significant benefit for improvement of symptom score and global assessment. Adverse events were not assessed as an outcome in this review. AUTHORS' CONCLUSIONS: There is no evidence that bulking agents are effective for treating IBS. There is evidence that antispasmodics are effective for the treatment of IBS. The individual subgroups which are effective include: cimetropium/dicyclomine, peppermint oil, pinaverium and trimebutine. There is good evidence that antidepressants are effective for the treatment of IBS. The subgroup analyses for SSRIs and TCAs are unequivocal and their effectiveness may depend on the individual patient. Future research should use rigorous methodology and valid outcome measures.